HLD - 101

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Agenda

- Intro
- Life of a packet
- DNS
- Scaling

x

HLD vs LLD

LLD
↳ code

HLD

Spec

↳ HLD

↓

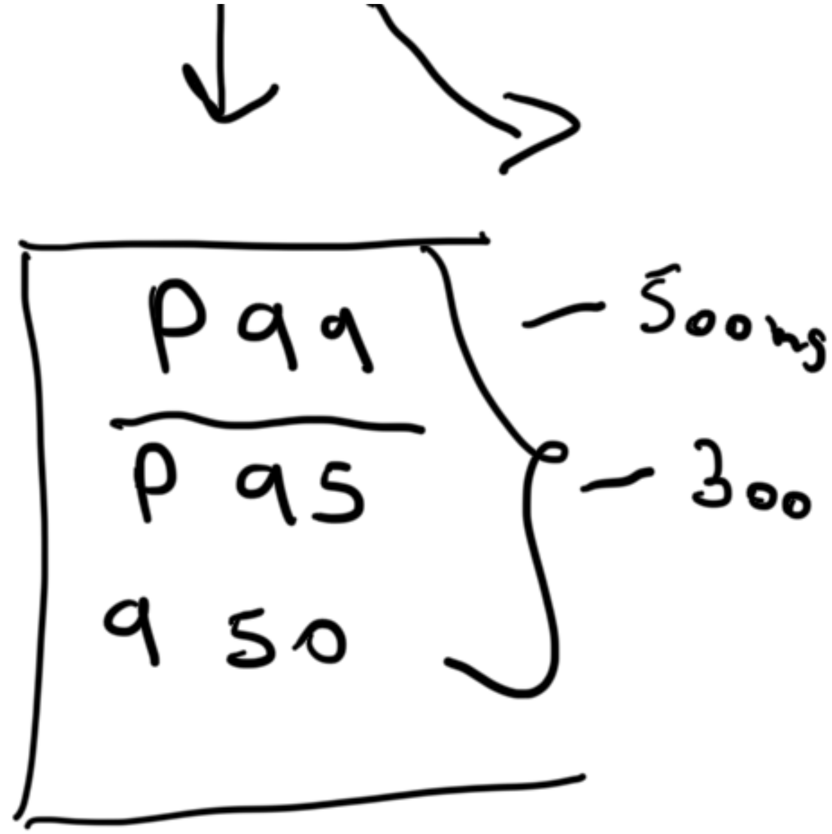
LLD

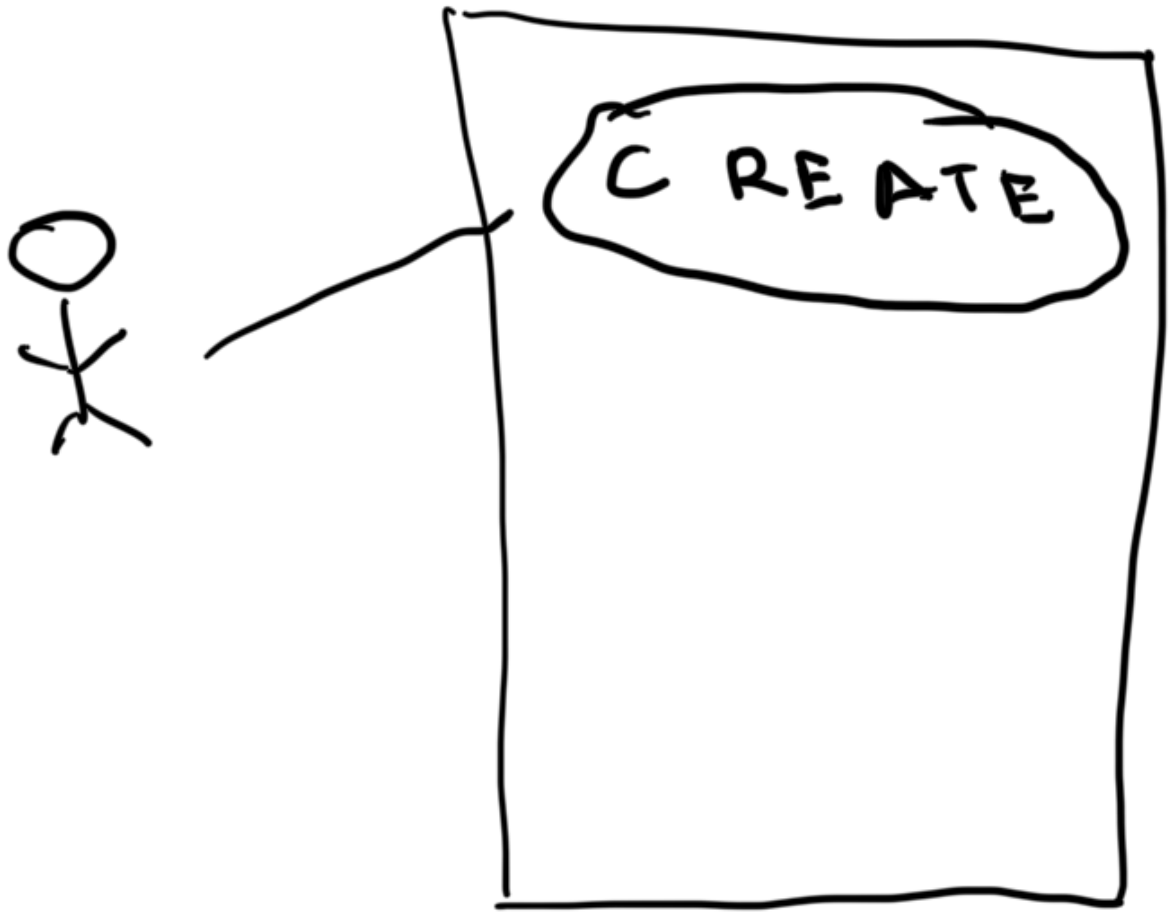
TD | ADR

Case study

Delicious

- ELK
- Datadog
- Prometheus
- micrometer





Create a bookmark

– POST /bookmarks

Get a bookmark

- GET /bk/{id}
- GET /bk

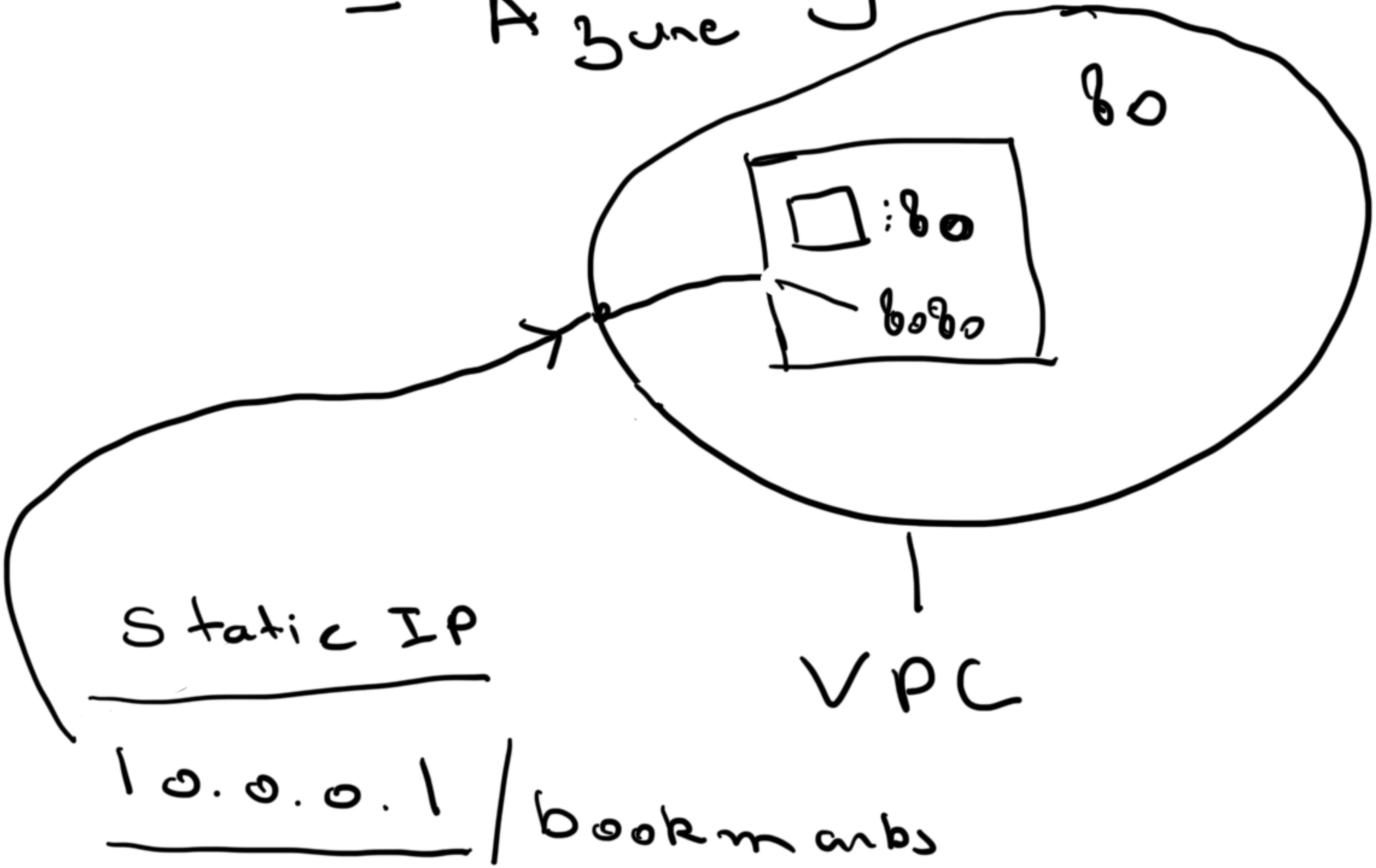
local - localhost

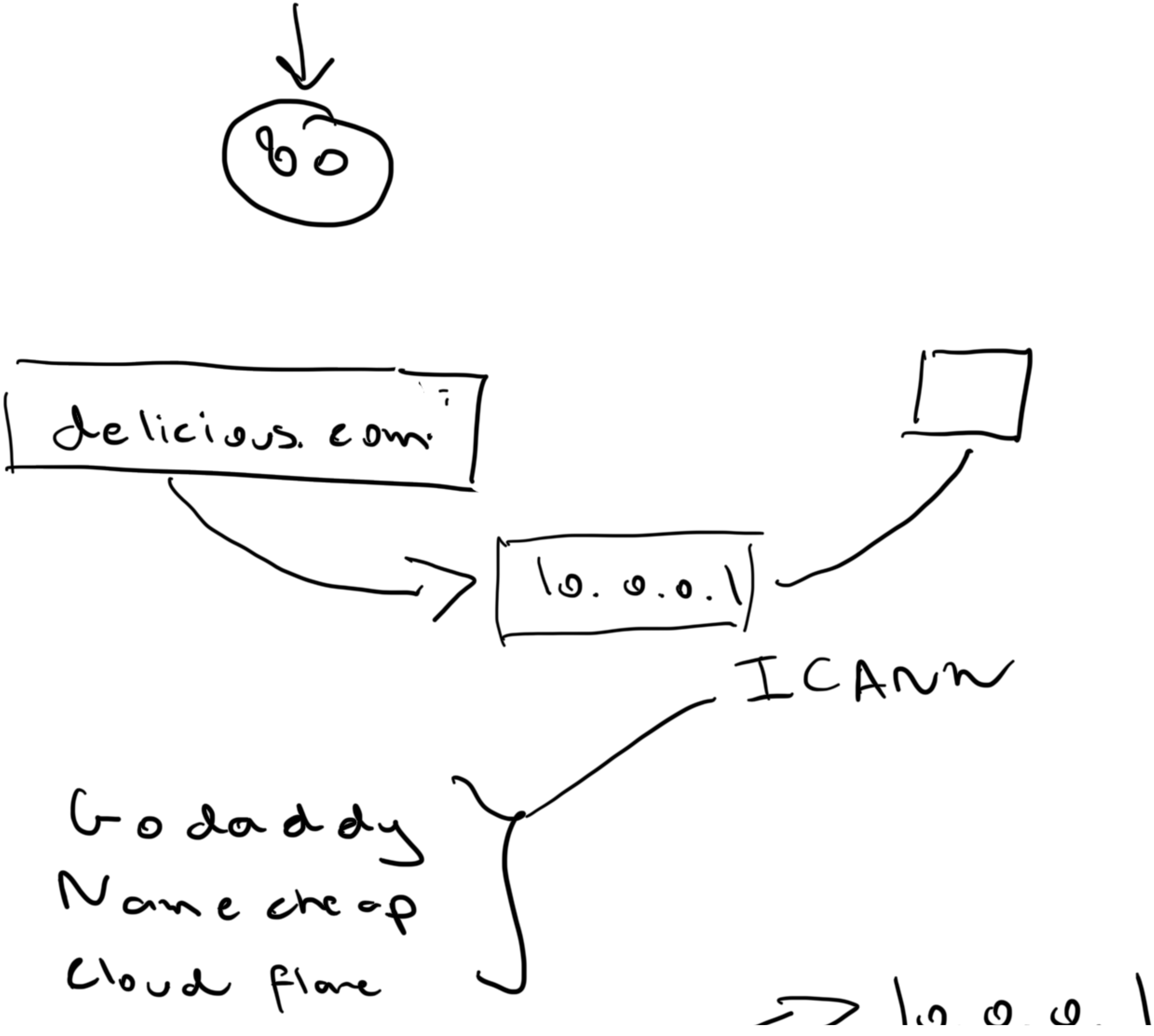
$\boxed{127.0.0.1:8080}$

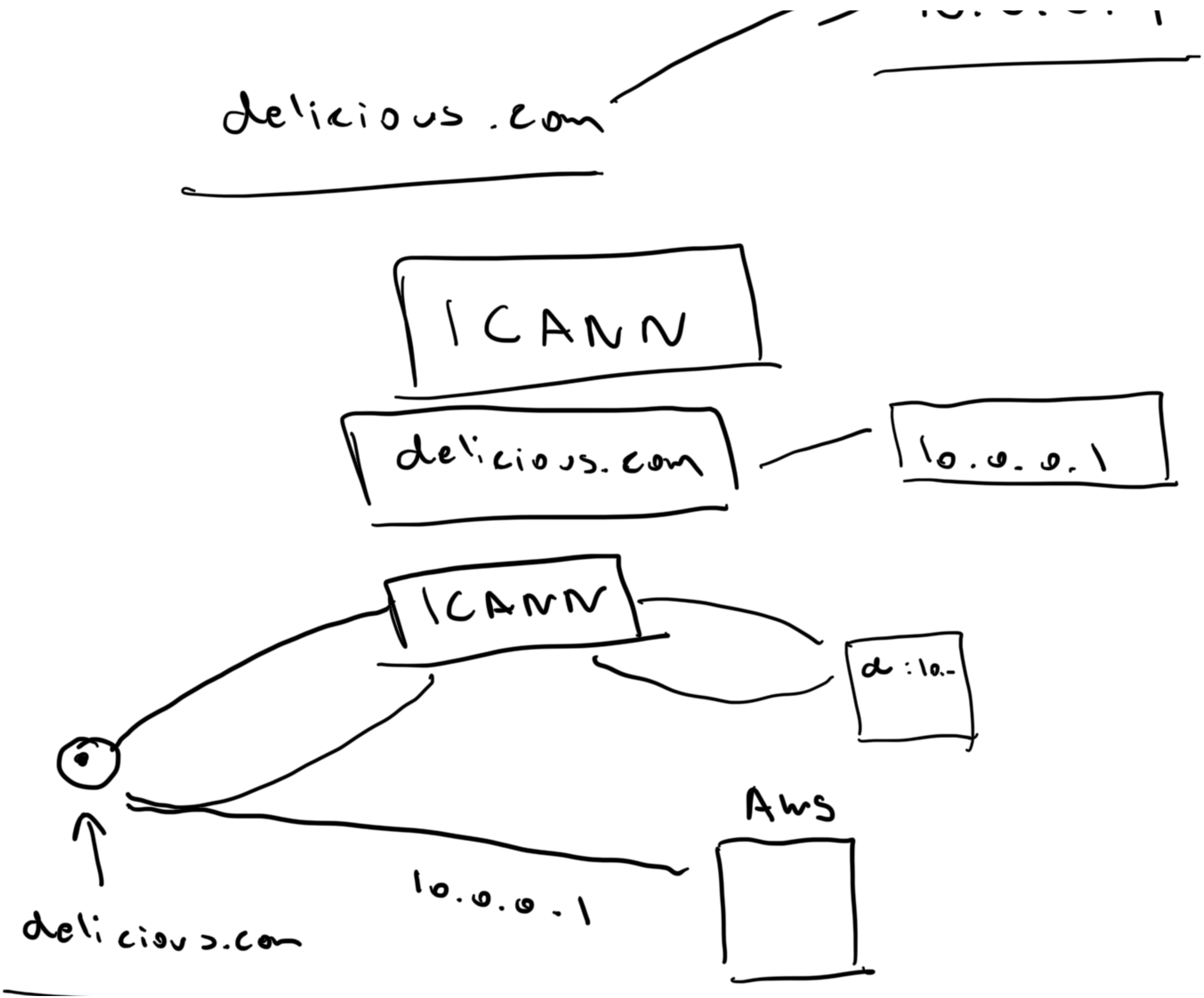
spring - 8080

Cloud providers

- AWS
- GCP
- Azure




80
delicious.com
10.0.0.1
ICANN
Godaddy
Namecheap
Cloud flare


delicious.com
ICANN
delicious.com
10.0.0.1
ICANN
AWS
10.0.0.1
delicious.com

Domain name resolution

- bottleneck | SPOF

Domain name system

(DNS)

1

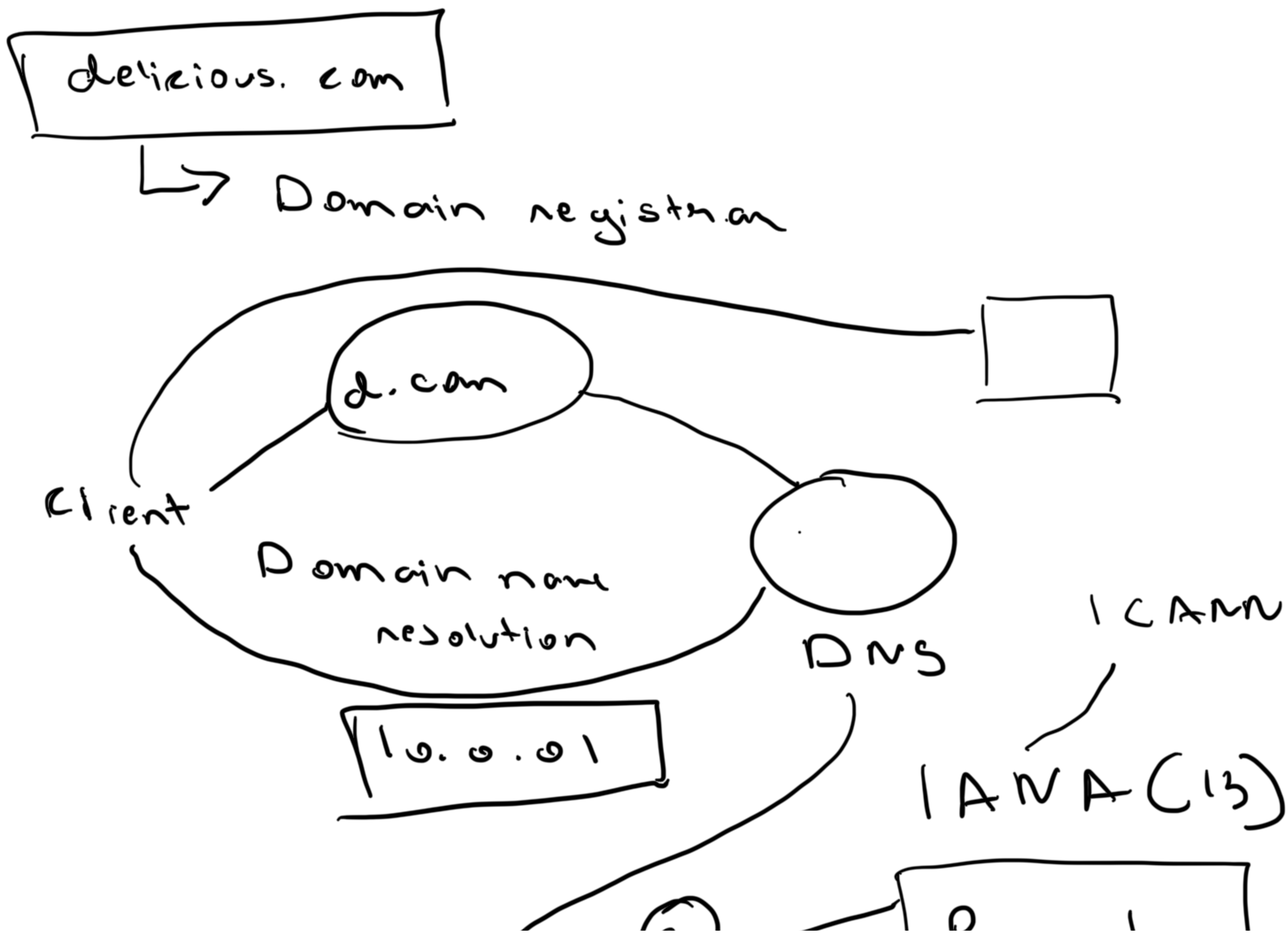
Browsers
Mobile
Postman

Clients

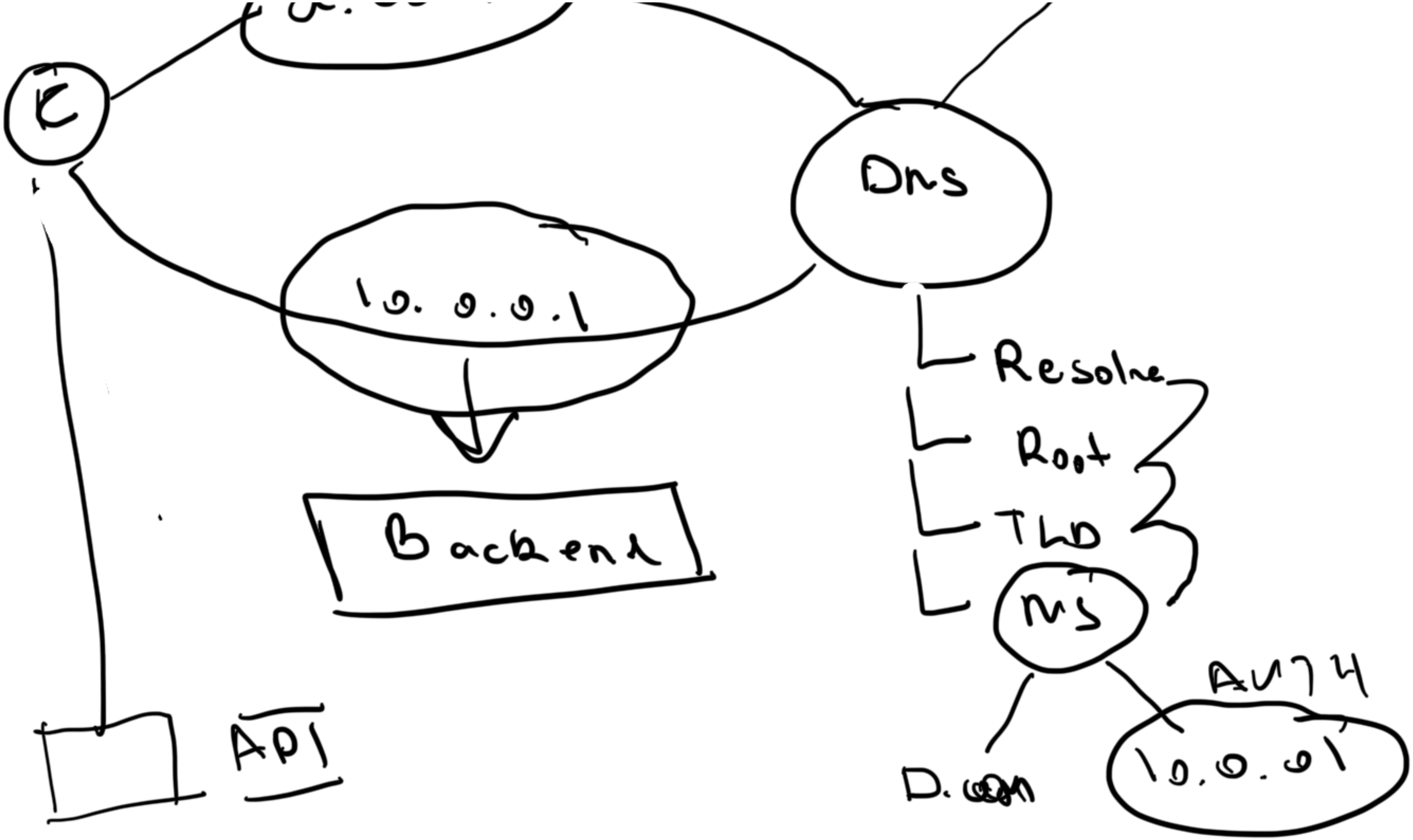
Break

$\boxed{6:10 — 6:20}$

- DNS
- Load balanu

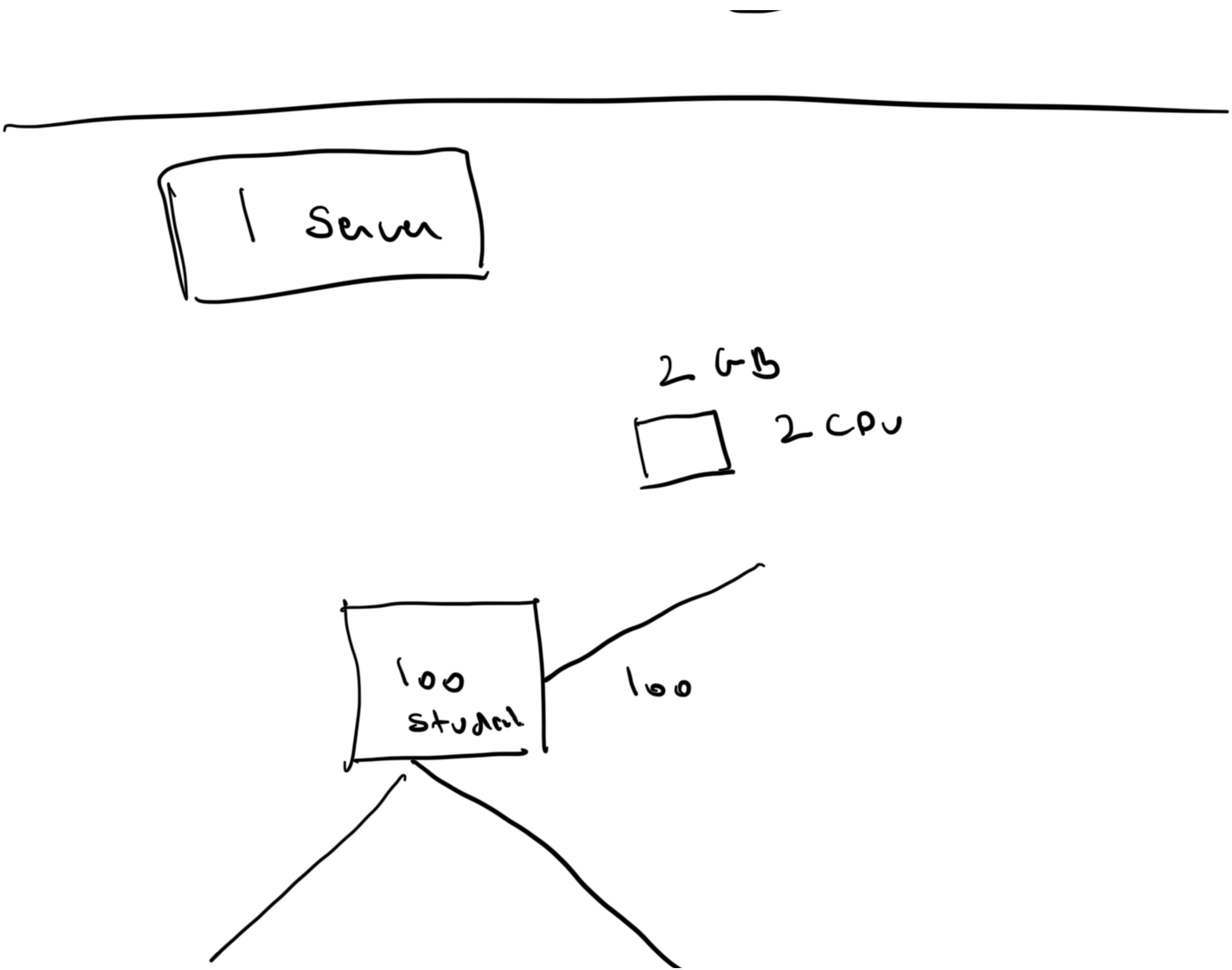

delicious. com
Domain registrar
d. com
Client
Domain name resolution
DNS
ICANN
10. 0. 01
IANA (13)

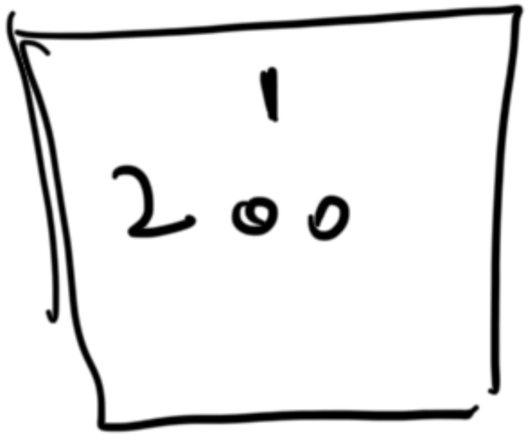
2
Root
3
.com
d.com
Resolver
d.com
Client
1
TLD
.com
.in
.org
4
- ISP
- Google
8.8.8.8
Name server
d.com - 10.0.0.1
.com
d.com



2009 — Dns

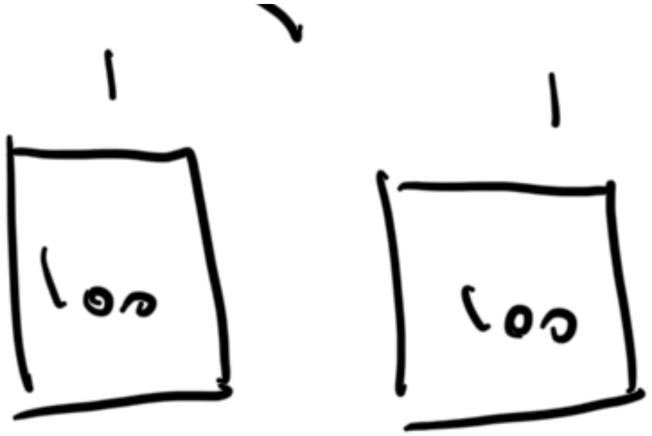
2026


1 Server
2 GB
2 CPU
100
student
100



Vertical

scaling up



Horizontal

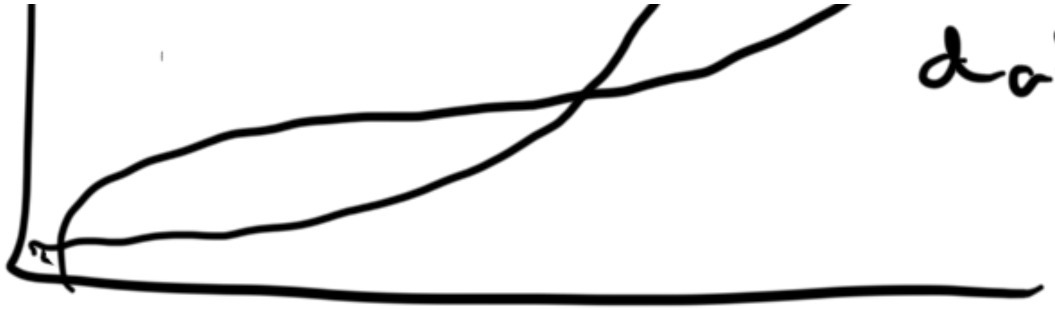
scaling out

Complexity

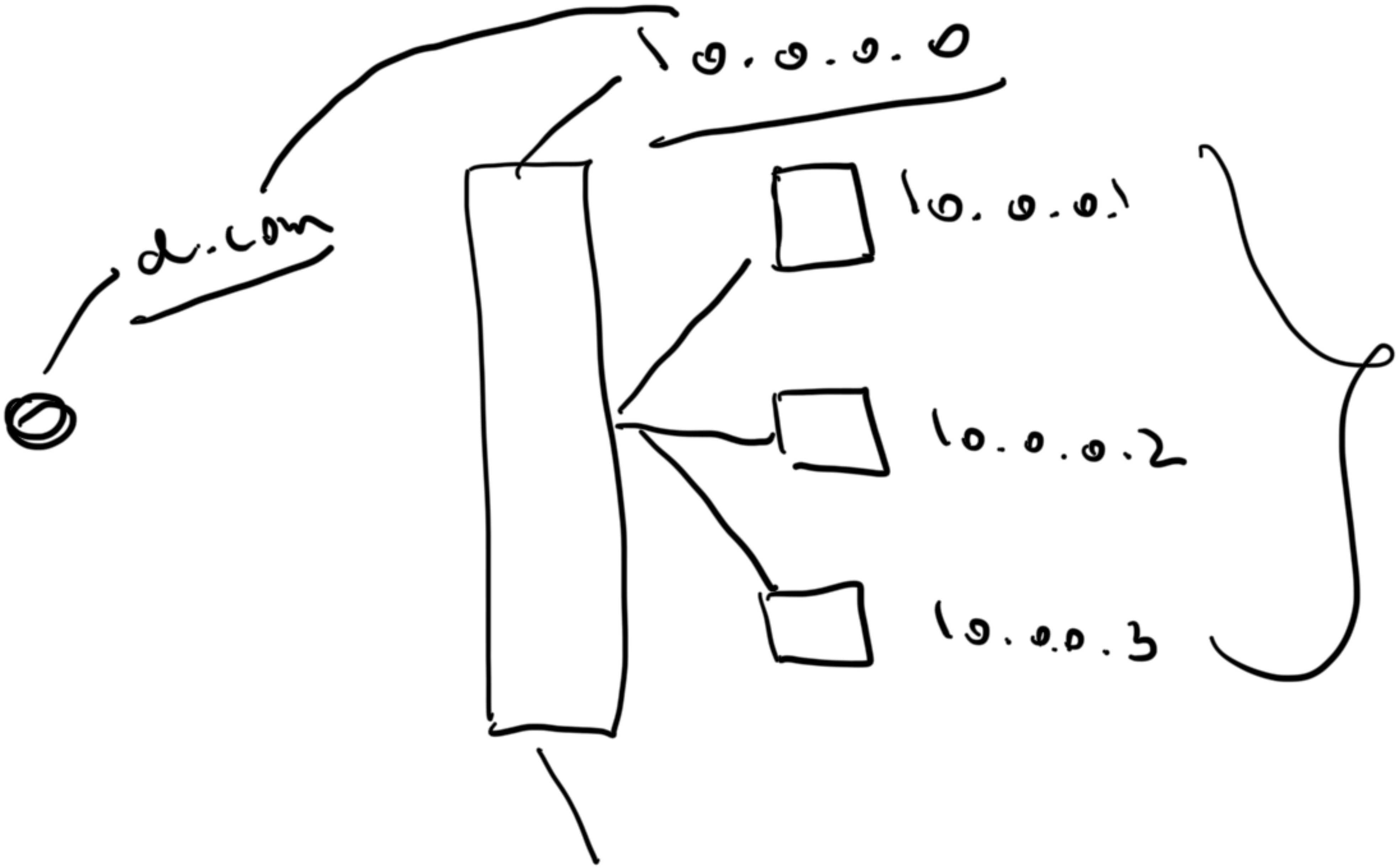
1. hard limit
2. Spof

app. servers — 4

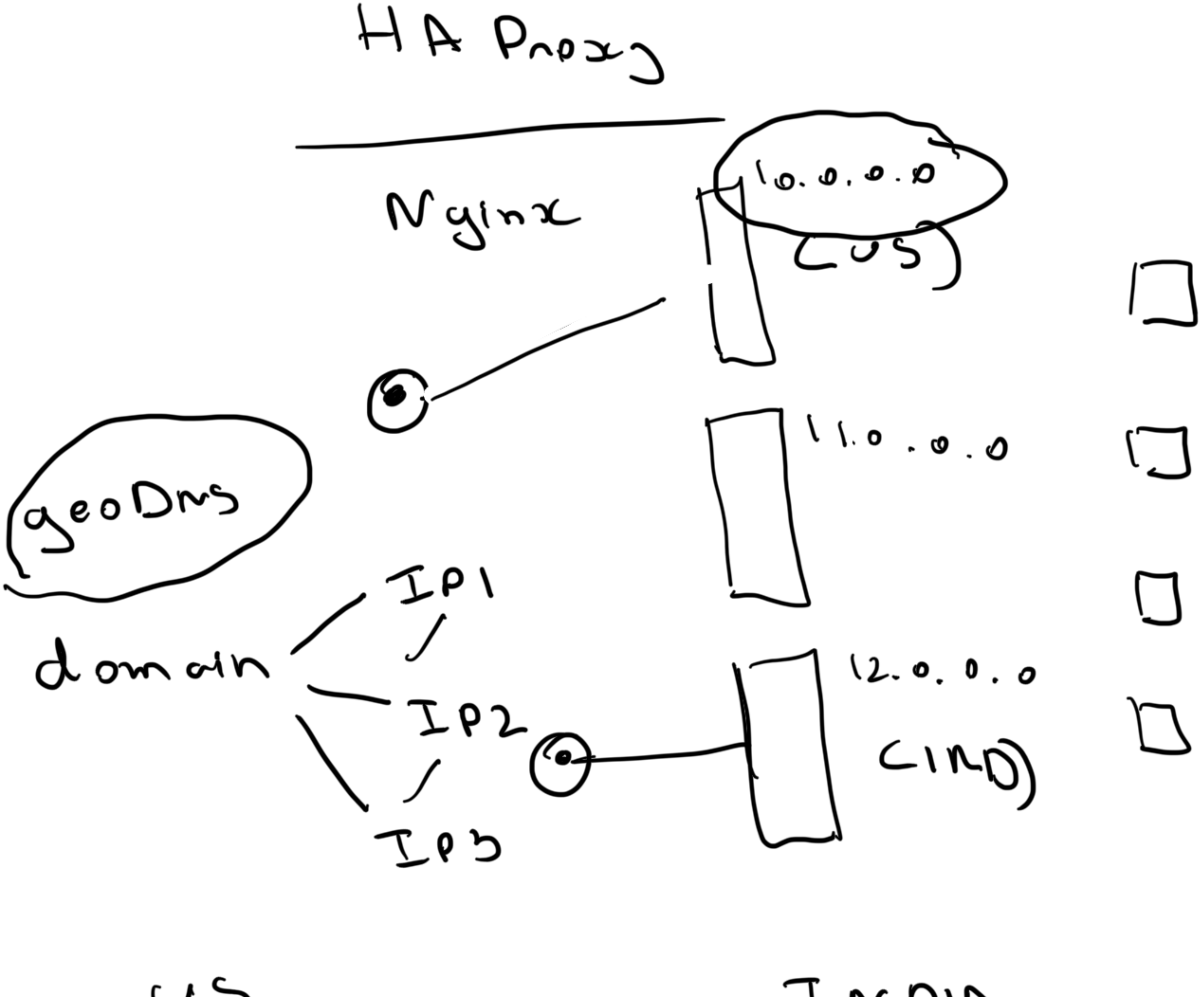
↳ API



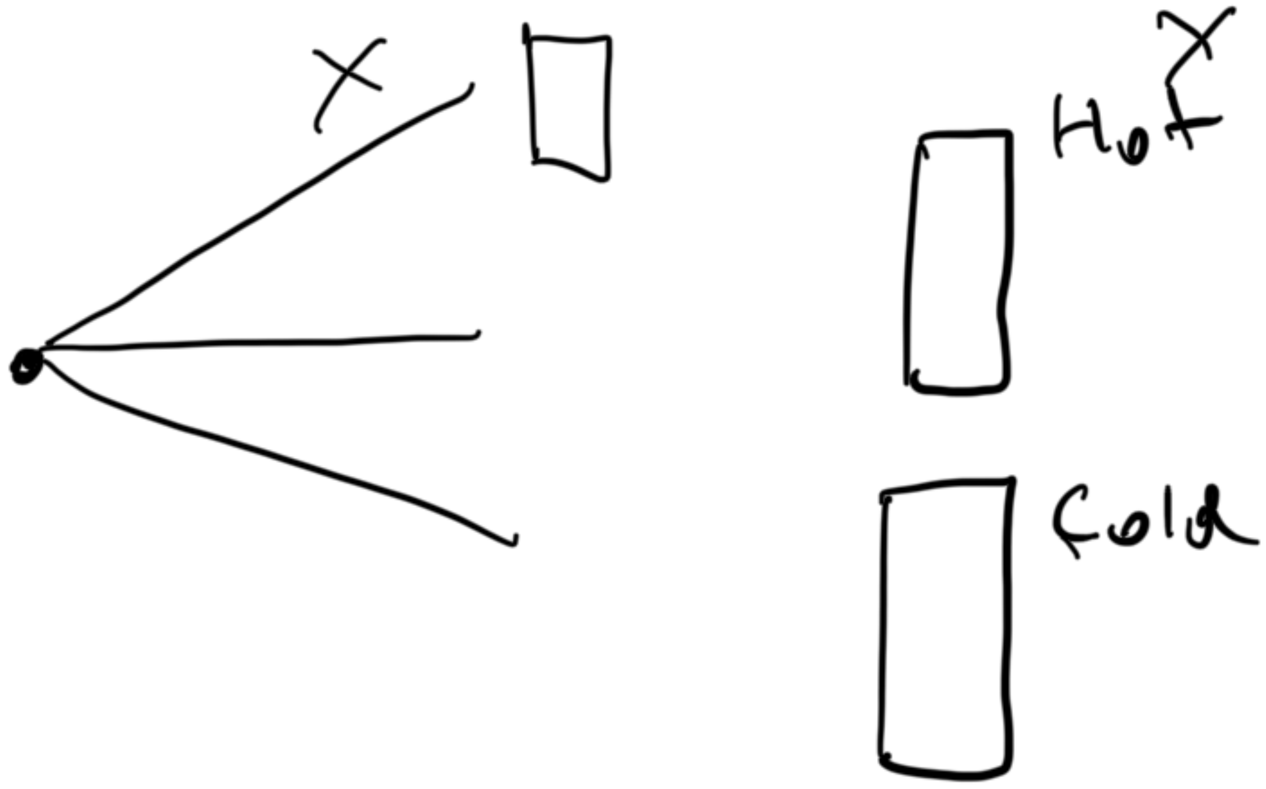
database
↳ vertical

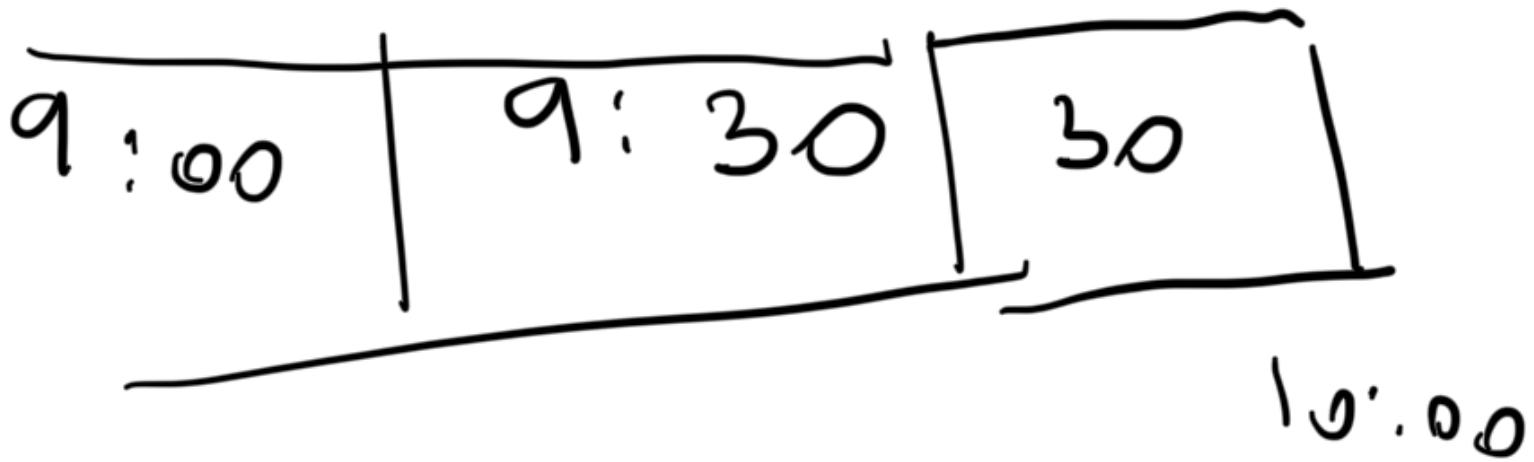


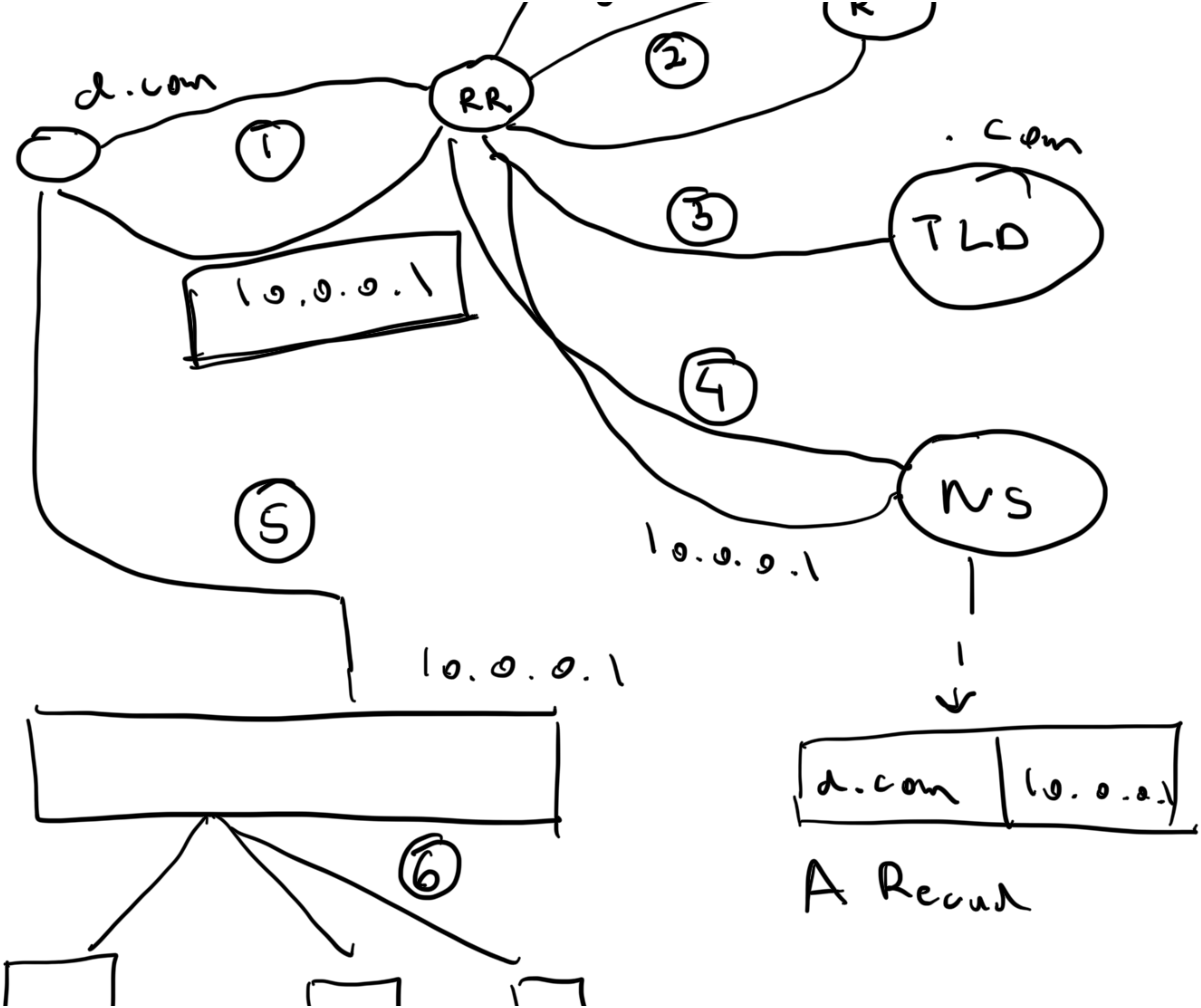
load balancer
(ingress)


HA Proxy
Nginx
10.0.0.0
(US)
11.0.0.0
12.0.0.0
(IND)
geoDns
domain
IP1
IP2
IP3

US INDIA

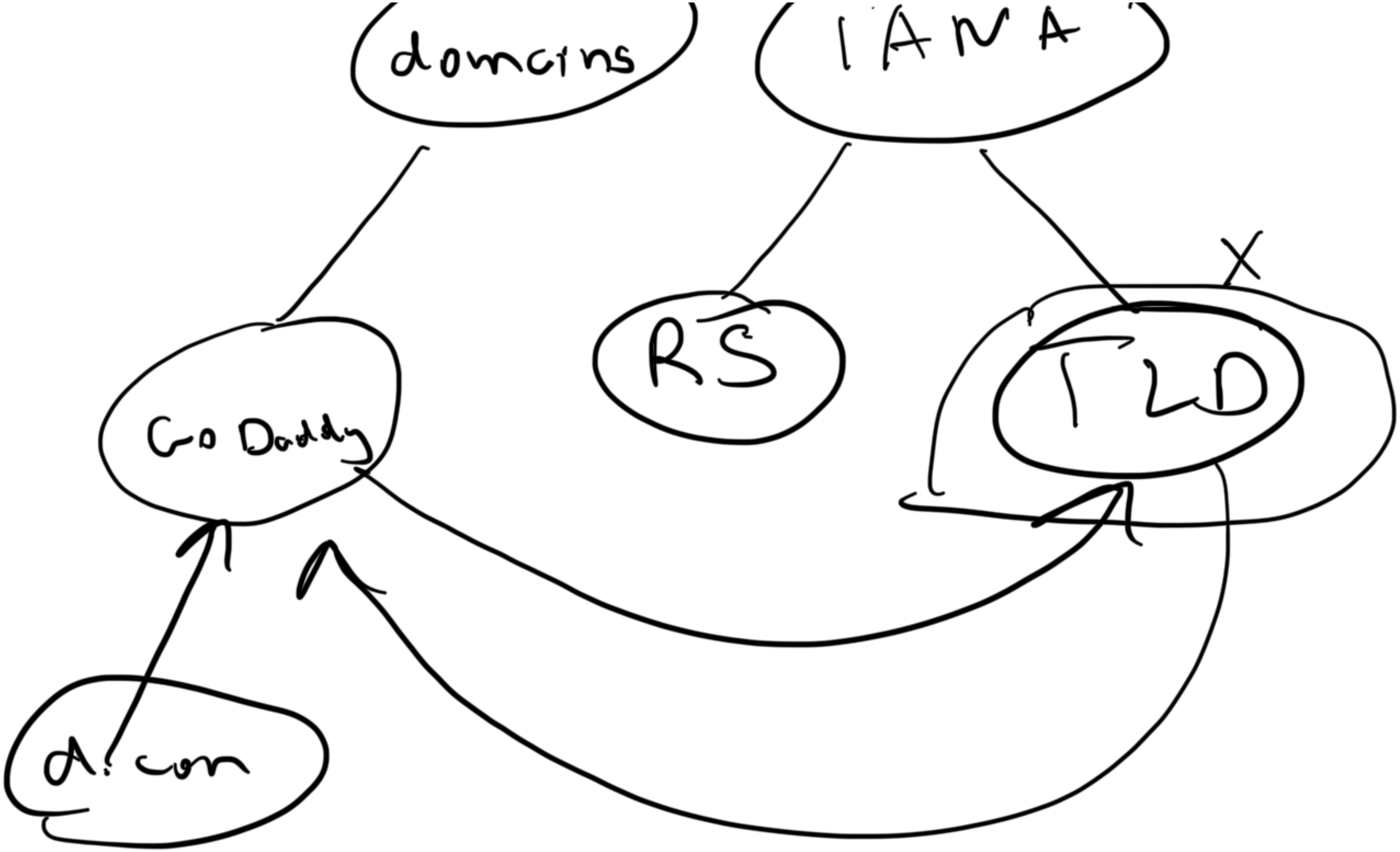




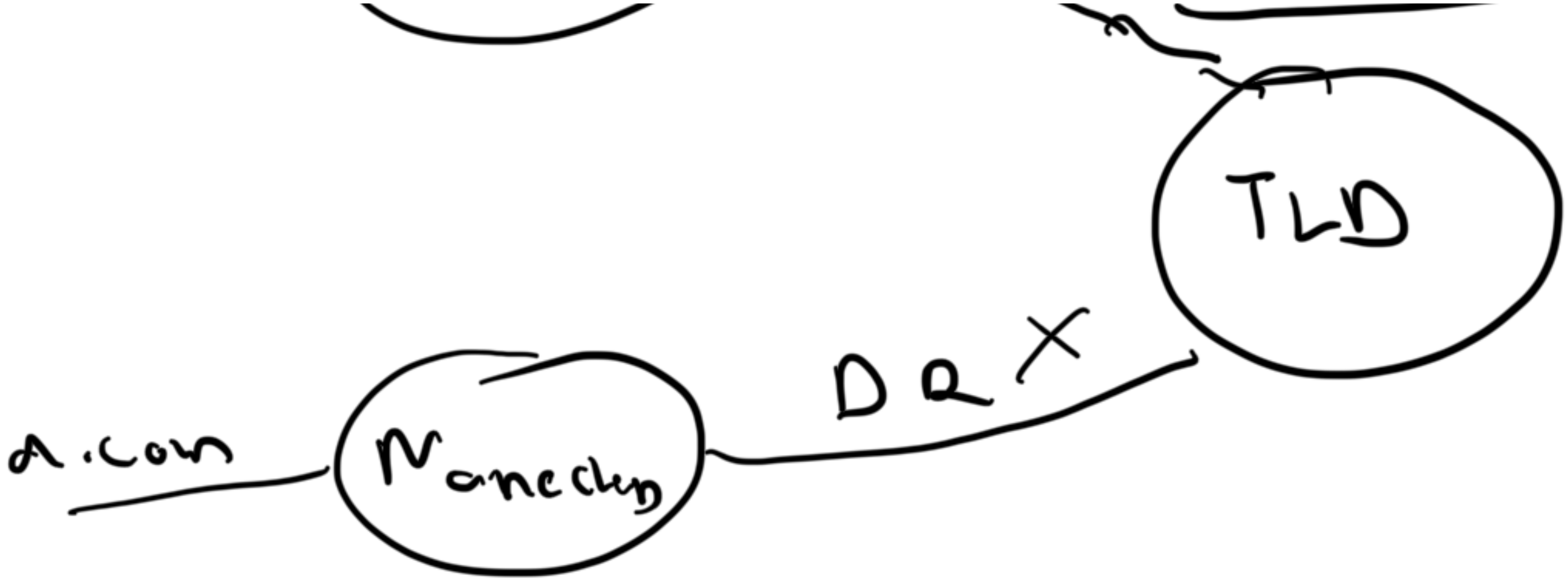


d.com
1
RR
2
R
.com
3
TLD
10.0.0.1
4
NS
10.0.0.1
5
10.0.0.1
d.com
10.0.0.1
A Record
6

1. Load balances
   - Load distribute
   - health check APIs
2. Sharding
   - Consistent hashing



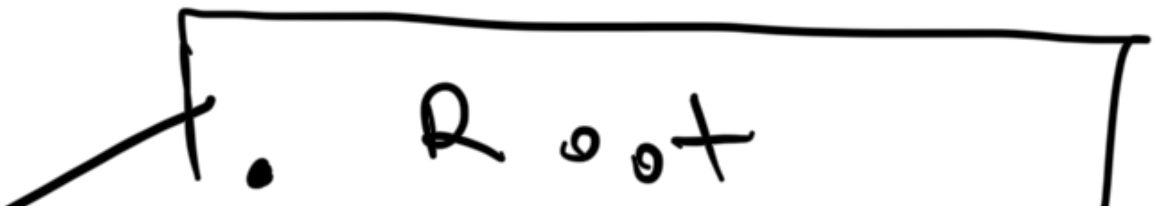
OR First one

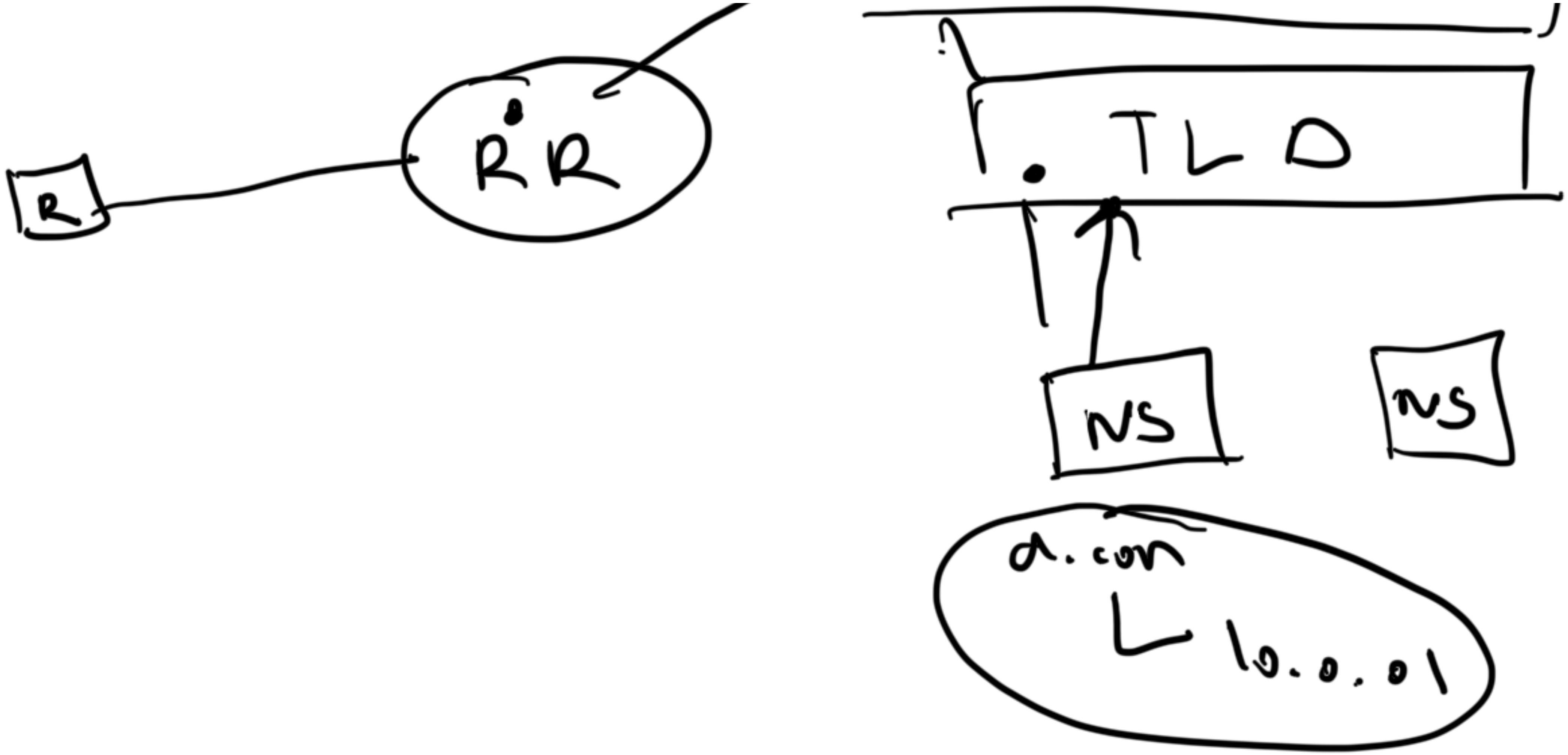


10:01 - LD a.com

10:02 - NC

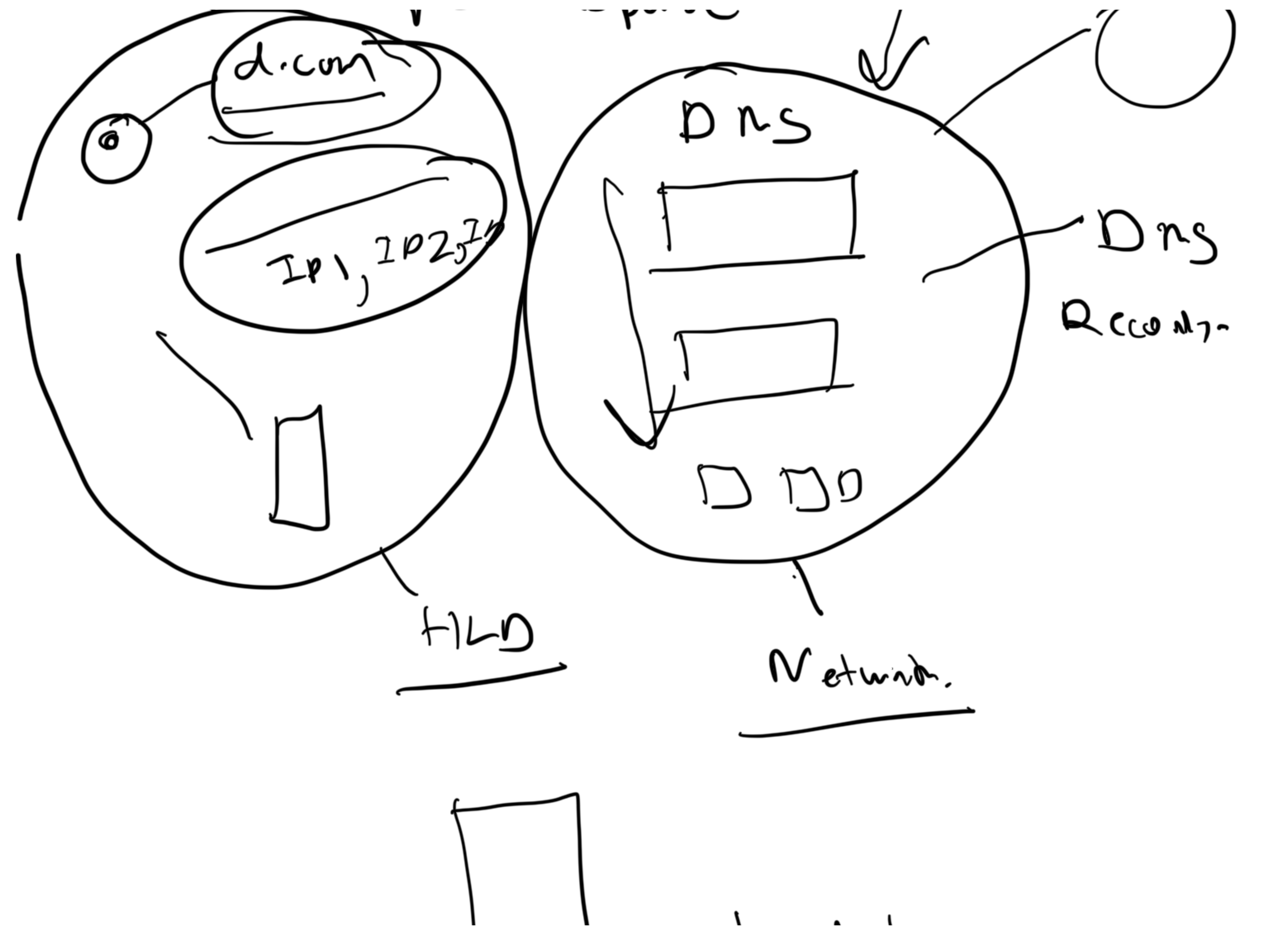
Registry DNS prop

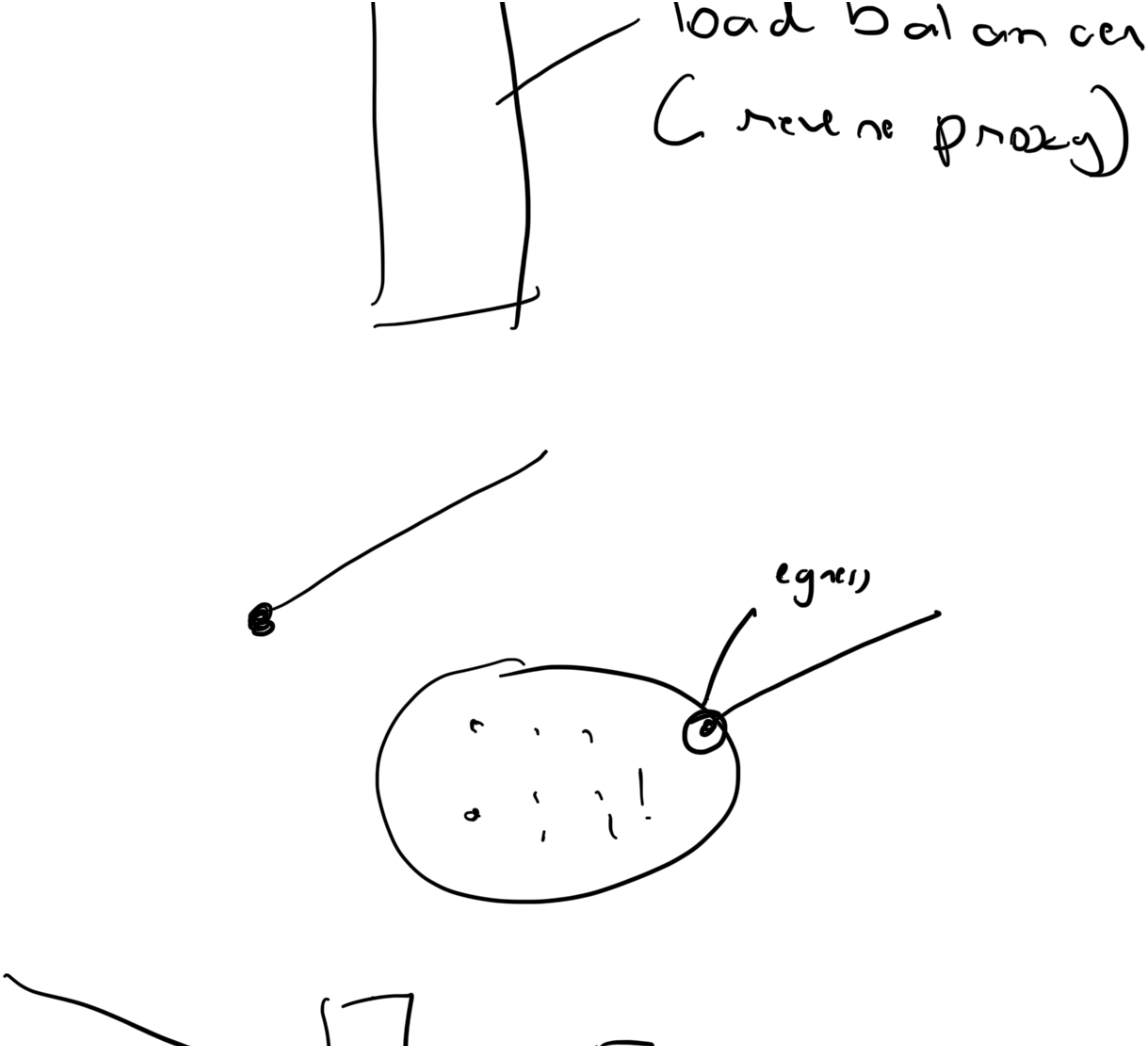




Domain name resolution

request -response


d.com
Ip1, IP2,
Dns
Dns
HLD
Network.


load balancer
(reverse proxy)
egress


ingress
reverse proxy
HTTP
load balancer
api gateway
nginx

TCP

bong